

MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Dashboard Guide:

Dashboards: Hospital Overview

Patient Flow

Department Analytics

Resource Utilization

1. Hospital Overview Dashboard

Purpose: The Hospital Overview Dashboard provides a high-level view of hospital performance and operations. It helps management understand admission trends, bed utilisation, occupancy, readmission levels, and department efficiency.

Filters Used:

- **Admission Month:** Used to view hospital performance for a selected month or time period.
- **Admission Type:** Used to analyse different types of patient admissions.
- **Department Name:** Used to view the performance of a specific department.
- **Ward Type:** Used to compare admissions and resource usage across different ward types.
- **Age Group:** Used to analyse admissions and discharges based on patient age categories.

KPIs Used:

- **Total Admissions:** Shows the total number of patient admissions and helps measure hospital workload.
- **Occupancy Rate:** Shows the percentage of beds occupied and helps monitor bed availability.
- **Readmission Rate:** Measures the percentage of patients admitted again and helps identify possible quality-of-care issues.
- **Bed Utilisation Rate:** Shows how effectively hospital beds are being used.
- **Department Efficiency:** Helps compare the operational performance of different departments.

Charts Used and Their Purpose:

- **Monthly Admission Trend:** Used to identify admission patterns and changes over time.
- **Admissions by Admission Type:** Used to compare admission categories and understand patient inflow.
- **Admissions by Ward Type:** Used to analyse how patients are distributed across different ward types.
- **Admissions by Age Group:** Used to understand which age groups contribute most to hospital admissions.
- **Discharges by Age Group:** Used to compare the number of discharged patients across different age groups.
- **Total Revenue by Departments:** Used to compare revenue generated by each department and identify high-revenue departments.
- **Average LOS by Department:** Used to compare the average length of patient stay across departments and identify departments with longer or shorter stays.

KeyInsights:

The dashboard helps identify admission trends, understand patient demographics, monitor occupancy and bed utilisation, compare department revenue, and analyse average patient stay. These insights support better resource planning, patient-flow management, and operational decision-making.

2. Patient Flow Dashboard

Purpose:

The Patient Flow Dashboard analyses the movement of patients through the hospital, from admission to discharge. It helps management understand patient volume, discharge trends, department-wise patient distribution, disease patterns, ward utilisation, and readmission levels.

Filters Used:

- **Admission Type:** Used to analyse elective and emergency admissions.
- **Department Name:** Used to view patient flow for a particular department.
- **Ward Type:** Used to compare patient distribution across different ward types.
- **Age Group:** Used to analyse patient flow across different age categories.
- **Blood Group:** Used to filter patients based on their blood group.

KPIs Used:

- **Total Patients:** Shows the number of unique patients treated in the hospital.
- **Total Discharges:** Shows the number of discharged patients.
- **Average Length of Stay:** Shows the average number of days patients stay in the hospital.
- **Readmission Rate:** Measures the percentage of patients admitted again.
- **Emergency Admission Rate:** Shows the percentage of admissions that were emergency cases.

Charts Used and Their Purpose:

- **Discharges by Monthly Trend:** Used to identify monthly discharge patterns and changes over time.
- **Patients by Department:** Used to compare the number of patients across departments.
- **Patients by Disease Category:** Used to understand the distribution of patients based on disease categories.
- **Patients by Ward Type:** Used to compare patient distribution across General, ICU, Private, and Semi-Private wards.
- **Admissions by Disease Name:** Used to identify the diseases contributing to patient admissions.
- **Discharges by Ward Type:** Used to compare discharge volumes across different ward types.

KeyInsights:

The dashboard helps identify patient admission and discharge patterns, departments with higher patient volumes, common disease categories, and ward-wise patient distribution. It also supports monitoring of average patient stay, emergency admissions, and readmission levels for better patient-flow management.

3. Department Analytics Dashboard

Purpose:

The Department Analytics Dashboard compares the performance of different hospital departments. It

helps management evaluate department-wise patient volume, efficiency, revenue, readmission rates, and resource requirements.

Filters Used:

- **Discharge Date:** Used to analyse department performance for a selected discharge period.
- **Disease Category:** Used to filter department analysis based on disease categories.
- **Ward Type:** Used to compare department performance across different ward types.
- **Disease Name:** Used to analyse departments based on specific diseases.

KPIs Used:

- **Total Departments:** Shows the number of departments included in the analysis.
- **Total Patient Payable:** Shows the total amount payable by patients.
- **Total Insurance Covered:** Shows the total amount covered by insurance.
- **Total Beds:** Shows the total bed capacity associated with the selected departments.
- **Total Revenue:** Shows the total revenue generated by the hospital.

Charts Used and Their Purpose:

- **Department Patients by Disease Category:** Used to compare patient counts across departments and disease categories.
- **Efficiency Score by Department:** Used to compare the operational efficiency of different departments.
- **Monthly Admissions by Department:** Used to analyse admission trends for each department over time.
- **Discharges by Department:** Used to compare the number of discharged patients across departments.
- **Readmission Rate by Department:** Used to identify departments with higher or lower readmission rates.
- **Patients Payable by Department:** Used to compare the patient-payable amount generated by each department.

Key Insights:

The dashboard helps identify departments with higher patient volumes, better efficiency scores, greater revenue contribution, and higher readmission rates. It also supports department-level resource planning, financial analysis, and performance improvement.

4. Resource Utilisation Dashboard

Purpose:

The Resource Utilisation Dashboard evaluates how effectively hospital resources such as beds, wards, and staff are being used. It helps management monitor resource availability, bed utilisation, staff allocation, and department-wise resource requirements.

Filters Used:

- **Admission Type:** Used to analyse resource utilisation based on admission categories.
- **Age Group:** Used to understand resource usage across different patient age groups.
- **Ward Type:** Used to compare resource utilisation across different ward types.

- **Department Name:** Used to analyse resource usage for a specific department.
- **Disease Category:** Used to examine resource utilisation based on disease categories.

KPIs Used:

- **Total Beds:** Shows the total number of beds available in the hospital.
- **Total Staff:** Shows the total number of hospital employees.
- **Total Wards:** Shows the total number of wards.
- **Average Beds per Ward:** Shows the average number of beds available in each ward.
- **Bed Utilisation Rate:** Measures how effectively hospital beds are being used.

Charts Used and Their Purpose:

- **Discharges by Disease Category:** Used to analyse discharge volumes across different disease categories.
- **Total Beds by Department:** Used to compare bed capacity across departments.
- **Beds by Ward Type:** Used to understand how beds are distributed among different ward types.
- **Average Staff Allocation by Department:** Used to compare staff allocation across departments.
- **Bed Utilisation by Department:** Used to monitor bed utilisation trends for each department over time.
- **Department-wise Payable by Age Group:** Used to compare patient-payable amounts across departments and age groups.

Key Insights:

The dashboard helps identify departments and wards with higher bed capacity, compare staff allocation, monitor bed utilisation trends, and understand resource requirements based on patient demographics and disease categories. These insights support better resource planning, capacity management, and operational efficiency.